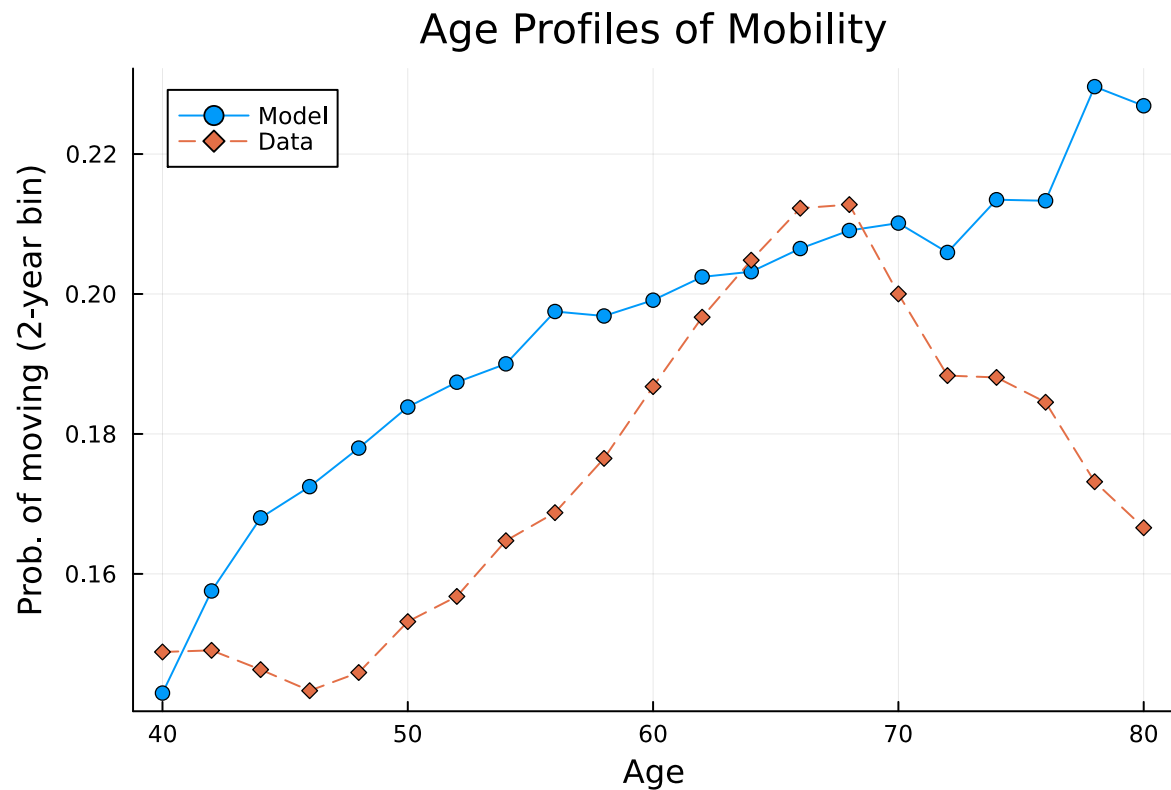


Graphing Migration Data and Model Estimates

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Graphs of Estimated and Data Probability of Migrating and Flows of Migration



The estimated age profiles showcase an interesting trend, basically making the growth in migration take place much earlier than in the data. The data shows that starting at around 50, people begin migrating more until peaking between 65-70. The model, however, is showing that people immediately start to migrate, and then it slows down at around 50-55, before getting

somewhat close to the data for the ages 60-70. No matter how much I fight it, the model wants the oldest members of the population to migrate the most, which goes against the data trend of a steep decline after 70, as well as the logic of disutility of moving ($\gamma(z')$ in the model) increasing with age. With that said, the model is capturing the general shape of the data, just shifted left, excluding the oldest ages.

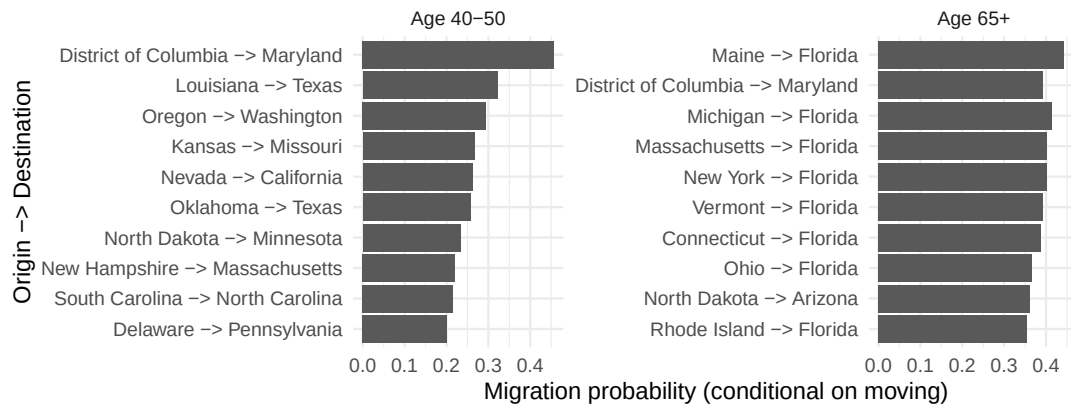
A tibble: 10 x 7

	age_group	origin	dest	origin_state	dest_state	flow_label	prob
	<chr>	<dbl>	<dbl>	<chr>	<chr>	<chr>	<dbl>
1	Age 40-50	11	24	District of Columbia	Maryland	District of ~	0.456
2	Age 40-50	22	48	Louisiana	Texas	Louisiana → ~	0.322
3	Age 40-50	41	53	Oregon	Washington	Oregon → Wa~	0.294
4	Age 40-50	20	29	Kansas	Missouri	Kansas → Mi~	0.267
5	Age 40-50	32	6	Nevada	California	Nevada → Ca~	0.262
6	Age 40-50	40	48	Oklahoma	Texas	Oklahoma → ~	0.257
7	Age 40-50	38	27	North Dakota	Minnesota	North Dakot~	0.234
8	Age 40-50	33	25	New Hampshire	Massachusetts	New Hampshi~	0.220
9	Age 40-50	45	37	South Carolina	North Carolina	South Carol~	0.216
10	Age 40-50	10	42	Delaware	Pennsylvania	Delaware → ~	0.202

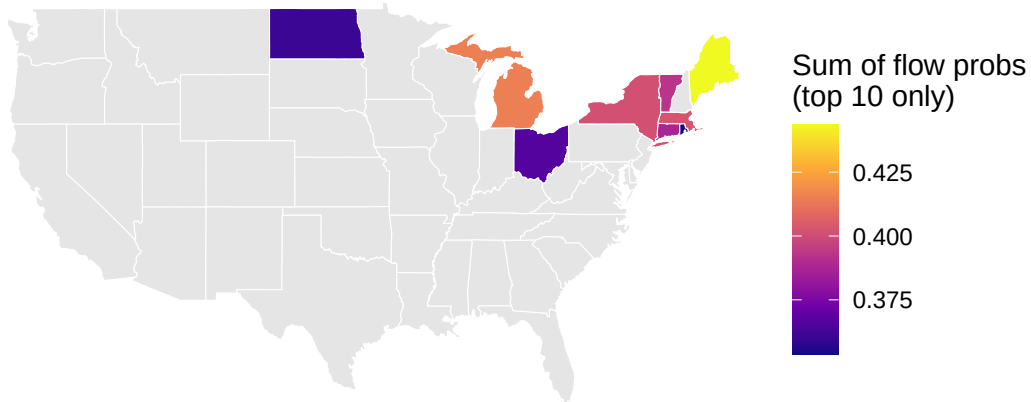
A tibble: 10 x 7

	age_group	origin	dest	origin_state	dest_state	flow_label	prob
	<chr>	<dbl>	<dbl>	<chr>	<chr>	<chr>	<dbl>
1	Age 65+	23	12	Maine	Florida	Maine → Florida	0.444
2	Age 65+	26	12	Michigan	Florida	Michigan → Flor~	0.415
3	Age 65+	25	12	Massachusetts	Florida	Massachusetts → ~	0.402
4	Age 65+	36	12	New York	Florida	New York → Flor~	0.401
5	Age 65+	11	24	District of Columbia	Maryland	District of Col~	0.394
6	Age 65+	50	12	Vermont	Florida	Vermont → Flori~	0.392
7	Age 65+	9	12	Connecticut	Florida	Connecticut → F~	0.387
8	Age 65+	39	12	Ohio	Florida	Ohio → Florida	0.366
9	Age 65+	38	4	North Dakota	Arizona	North Dakota → ~	0.361
10	Age 65+	44	12	Rhode Island	Florida	Rhode Island → ~	0.354

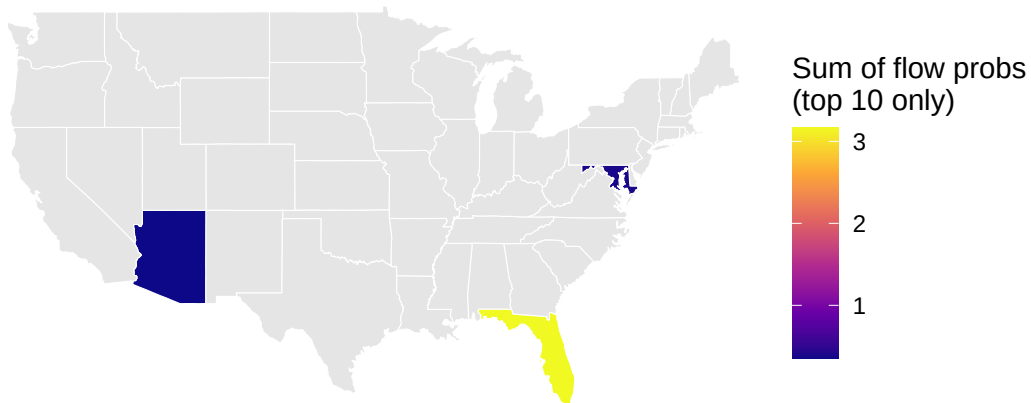
Top 10 Migration Flows by Age Group



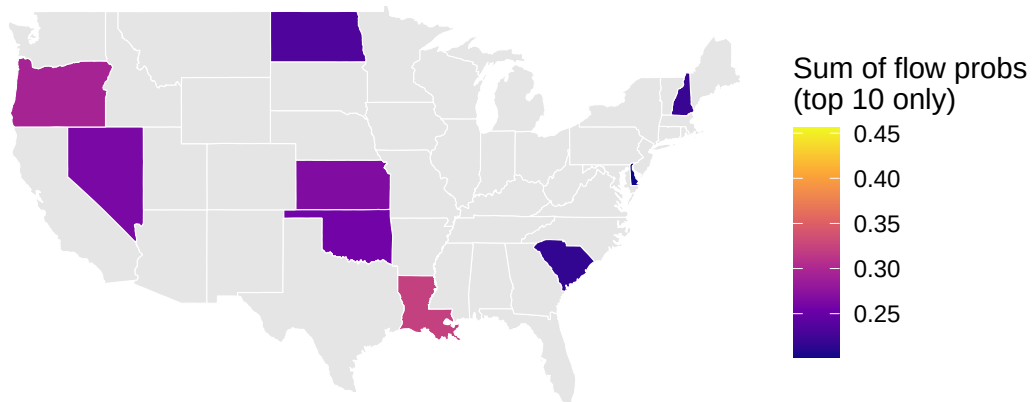
Top 10 65+ migration flows: origin intensity



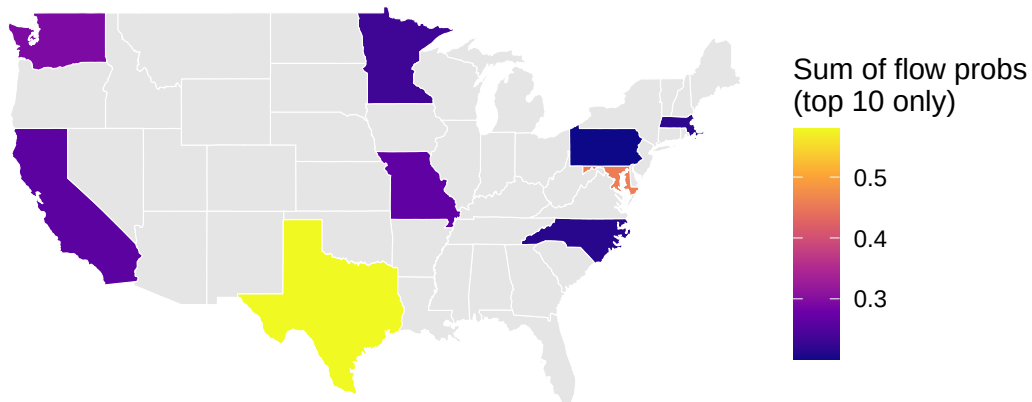
Top 10 65+ migration flows: destination intensity



Top 10 40–50 migration flows: origin intensity



Top 10 40–50 migration flows: destination intensity



These differ from previous flows and probabilities due to me correctly filtering for movers. Still see general trend of older people liking Florida, and workers liking states where they can still earn good wages.

Estimated Migration Flows

A tibble: 10 x 9

	origin_idx	dest_idx	origin_fips	dest_fips	origin_name	dest_name	prob
	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>	<dbl>
1	29	41	32	45	Nevada	South Carolina	0.0423
2	12	37	15	40	Hawaii	Oklahoma	0.0398
3	22	5	25	6	Massachusetts	California	0.0396
4	8	3	10	4	Delaware	Arizona	0.0393
5	20	1	23	1	Maine	Alabama	0.0379
6	38	4	41	5	Oregon	Arkansas	0.0375
7	32	37	35	40	New Mexico	Oklahoma	0.0371

```

      8      45      43      49      47 Utah      Tennessee      0.0368
      9      51      34      56      37 Wyoming      North Carolina 0.0366
     10      49      34      54      37 West Virginia North Carolina 0.0364
# i 2 more variables: age_group <chr>, flow_label <chr>

```

```
# A tibble: 10 x 9
```

	origin_idx <dbl>	dest_idx <dbl>	origin_fips <dbl>	dest_fips <dbl>	origin_name <chr>	dest_name <chr>	prob <dbl>
1	44	3	48	4	Texas	Arizona	0.0601
2	7	1	9	1	Connecticut	Alabama	0.0526
3	5	3	6	4	California	Arizona	0.0520
4	3	34	4	37	Arizona	North Carolina	0.0505
5	37	1	40	1	Oklahoma	Alabama	0.0501
6	3	37	4	40	Arizona	Oklahoma	0.0499
7	41	3	45	4	South Carolina	Arizona	0.0494
8	41	5	45	6	South Carolina	California	0.0494
9	4	5	5	6	Arkansas	California	0.0491
10	8	43	10	47	Delaware	Tennessee	0.0491

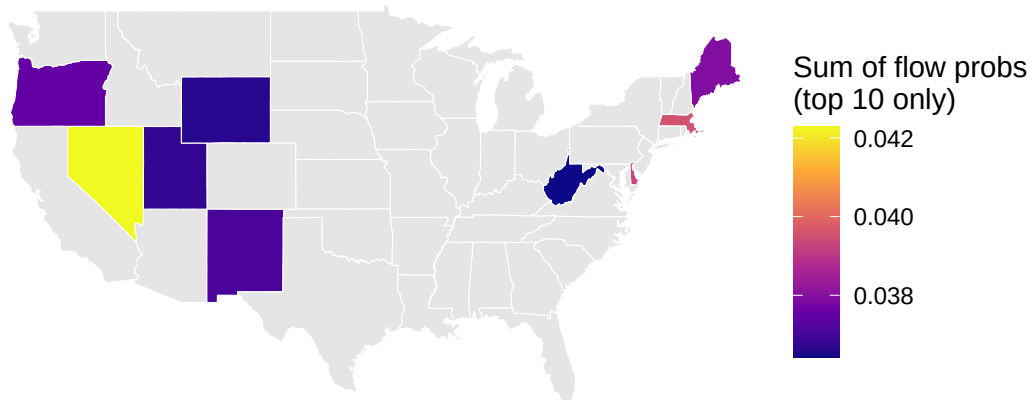
```

# i 2 more variables: age_group <chr>, flow_label <chr>

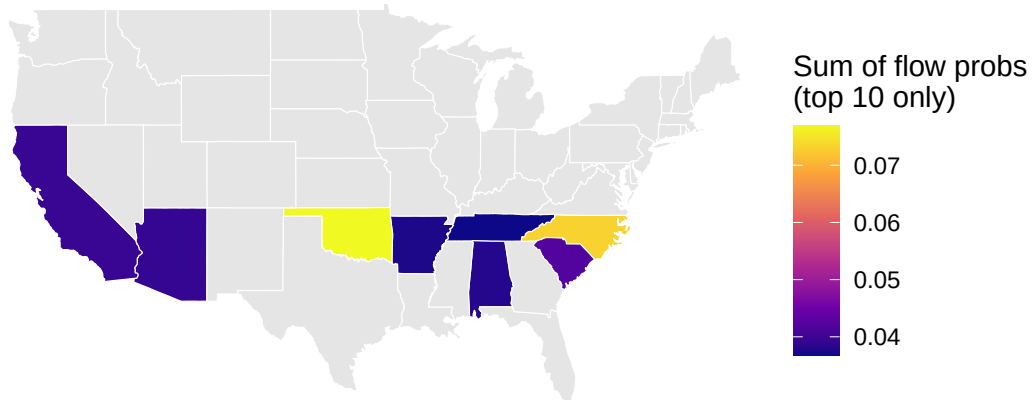
```

Low probabilities of migration flows compared to data. Unsure of why besides not able to capture all non-wage factors correctly.

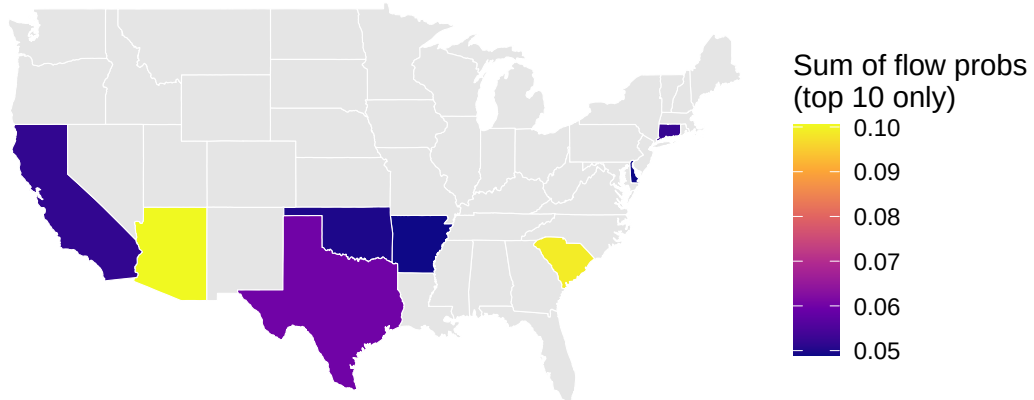
Top 10 model flows (ages 40–50): origin intensity



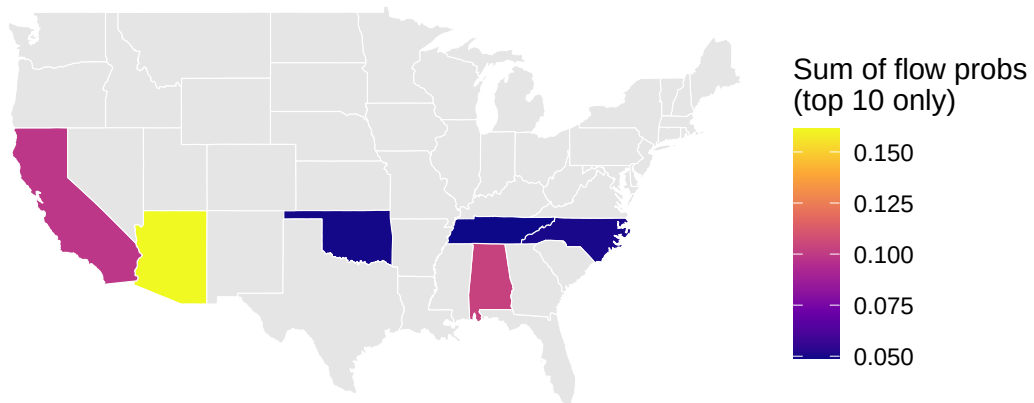
Top 10 model flows (ages 40–50): destination intensity



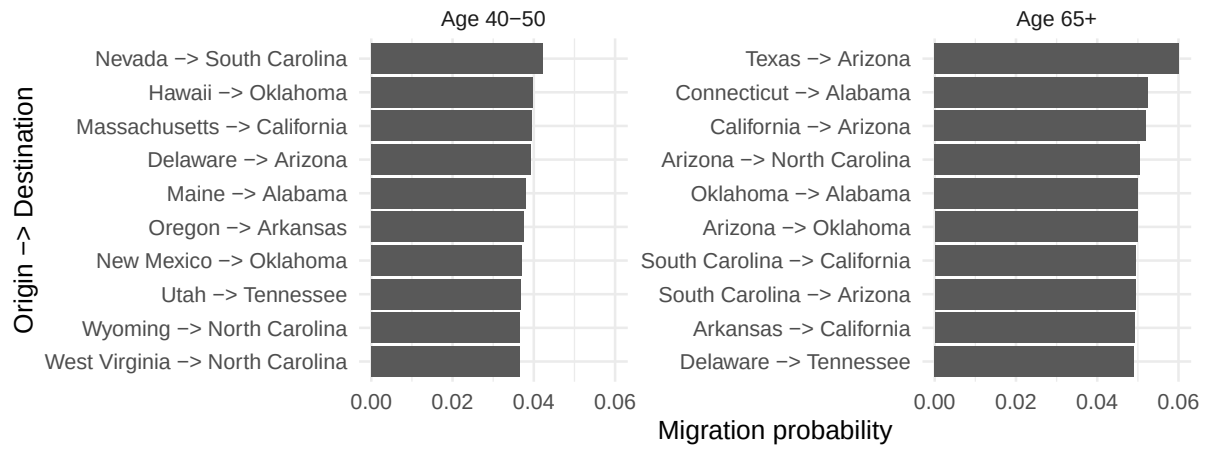
Top 10 model flows (ages 65+): origin intensity



Top 10 model flows (ages 65+): destination intensity



Top 10 Estimated Migration Flows by Age Group



Very different migration flow patterns in model vs data. Same as above.